

Implementation of Data Assimilation Research Testbed, DART, at CMCC

First results

Giovanni Conti, CMCC

CMCC

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CESM2+DART on ZEUS

- CESM2 (<http://www.cesm.ucar.edu/models/cesm2/>)

The compset used:

HIST_CAM60_CLM50%SP_CICE%PRES_DOCN%DOM_MOSART_SGLC_SWAV

It was the only compset tested at CMCC with CESM2. I've changed the SST in order to simulate 2017 and fixed the dusts and greenhouse gases at 2010

- DART Manhattan (<https://dart.ucar.edu>)

The only change was in `filter_mod.f90` (due to segmentation fault):

```
This ——> if (my_task == io_task) allocate(obs_temp(num_obs_in_set))  
substituted with:  
if (my_task==io_task) then  
    allocate(obs_temp(num_obs_in_set))  
else  
    (obs_temp(1))  
endif
```

We are using now intel 20.1 compilers (waiting for the “imminent” update 20.3)



CESM2+DART on ZEUS

In order to create a workaround for the issues related to

1. the Infiniband bugs (that brings mpi_finalization problems both for CESM2 and DART)
2. small cluster often overcrowded

we have created a small suite of scripts that decouple CESM2 and DART and allows for the creation and run of the experiments. The CESM2 members run in different jobs.

The suite is in a GitHub repository (https://github.com/giovanniconti83/suite_assimilation-atmo-CMCC)

Lots of files must be “cleaned” and some functionality like compression must be added. We hope that this could help also someone else in the future that has to work on a small intel cluster...



CESM2+DART on ZEUS

The Source Code Tree:

- `case_archive/` (Archive of the experiments templates)
- `main_scripts/` (Main scripts, e.g. create, cycle, assimilate and auxiliary txt files)
- `diagnostic/` (routines to diagnose assimilation performance)

The rationale behind

The workflow can be divided into three phases. There is a configurable script for every phase.

1. Phase1: Creation of the case template and of all the clones with links to the right I.C.s
(`main_scripts/cases_create.sh`)
2. Phase2: Run the experiment (`main_scripts/cases_cycle.sh`)
3. Phase3: Diagnostic (`diagnostic/cases_diag.sh`)



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Phase 1: CREATION

`cases_create.sh`

Variable to set:

<code>case_name:</code>	name of the experiment
<code>clonesroot:</code>	where the cases of the single members will be created
<code>nens:</code>	number of clones
<code>dartroot:</code>	DART directory
<code>tmpdir:</code>	auxiliary directory where the filter produces output
<code>archdir:</code>	archive dir

(To do: define which I.C. use for the first step. At the moment are fixed!)

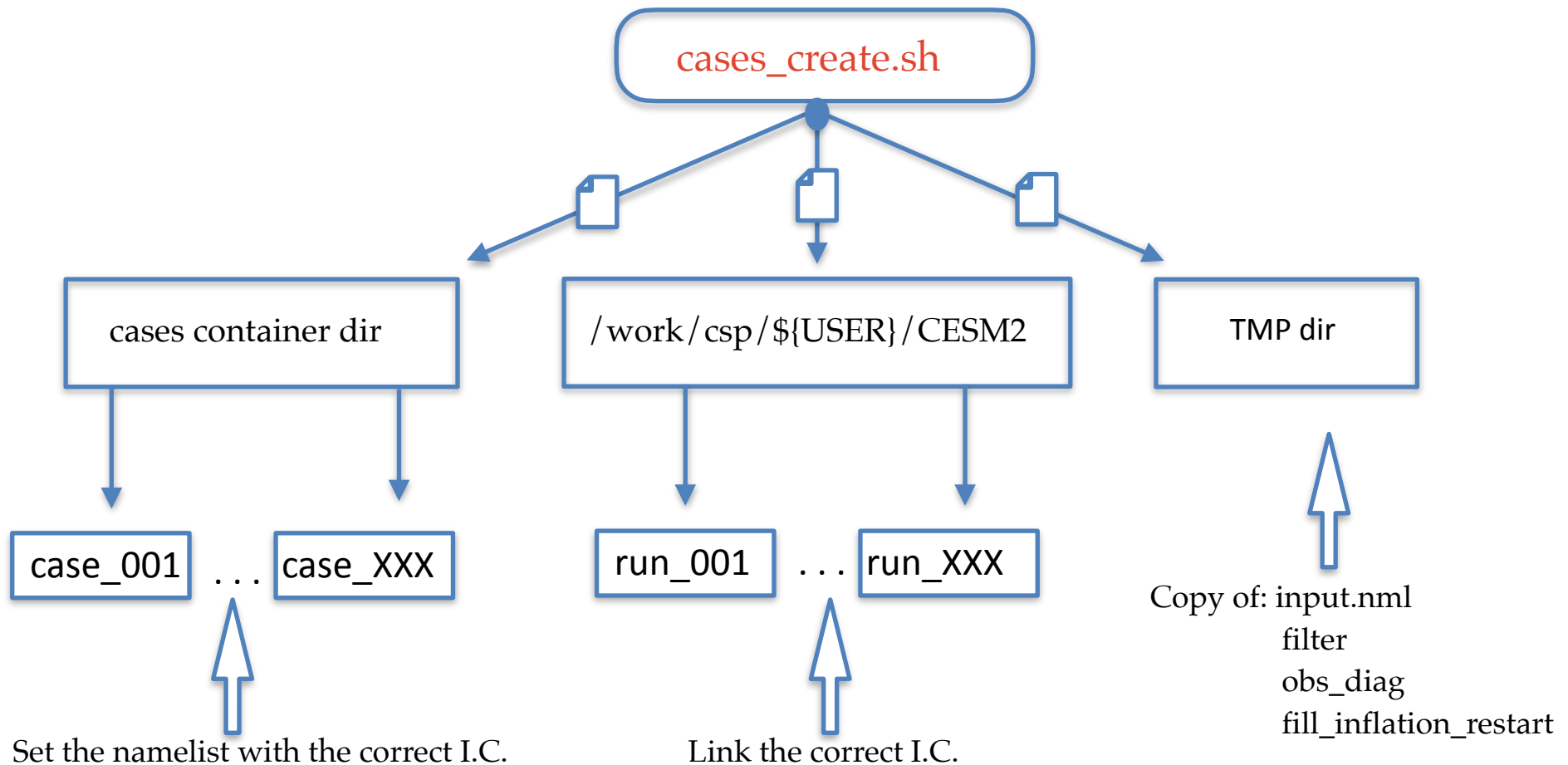
Dependencies:

```
cases_creates.sh  --> case.template.original  ( experiment template to clone )
                  |
                  --> input.nml.original      ( DART assimilation parameters )
```



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Phase 1: CREATION



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Phase 2: RUN THE EXPERIMENT

USAGE:>

```
nohup ./cases_cycle.sh > log.txt &
```

Variable to set:

NCYCLES:	number of CESM2 (and possibly DART) cycle
CONT_RUN:	TRUE if start from a previous run, FALSE for the first cycle
ACTIVATE_ASSI:	TRUE if you want to activate the assimilation (set automatically DATA_ASSIMILATION_ATM=TRUE)
MAXTRY:	number of attempts to make assimilation in case of failure

Dependencies:

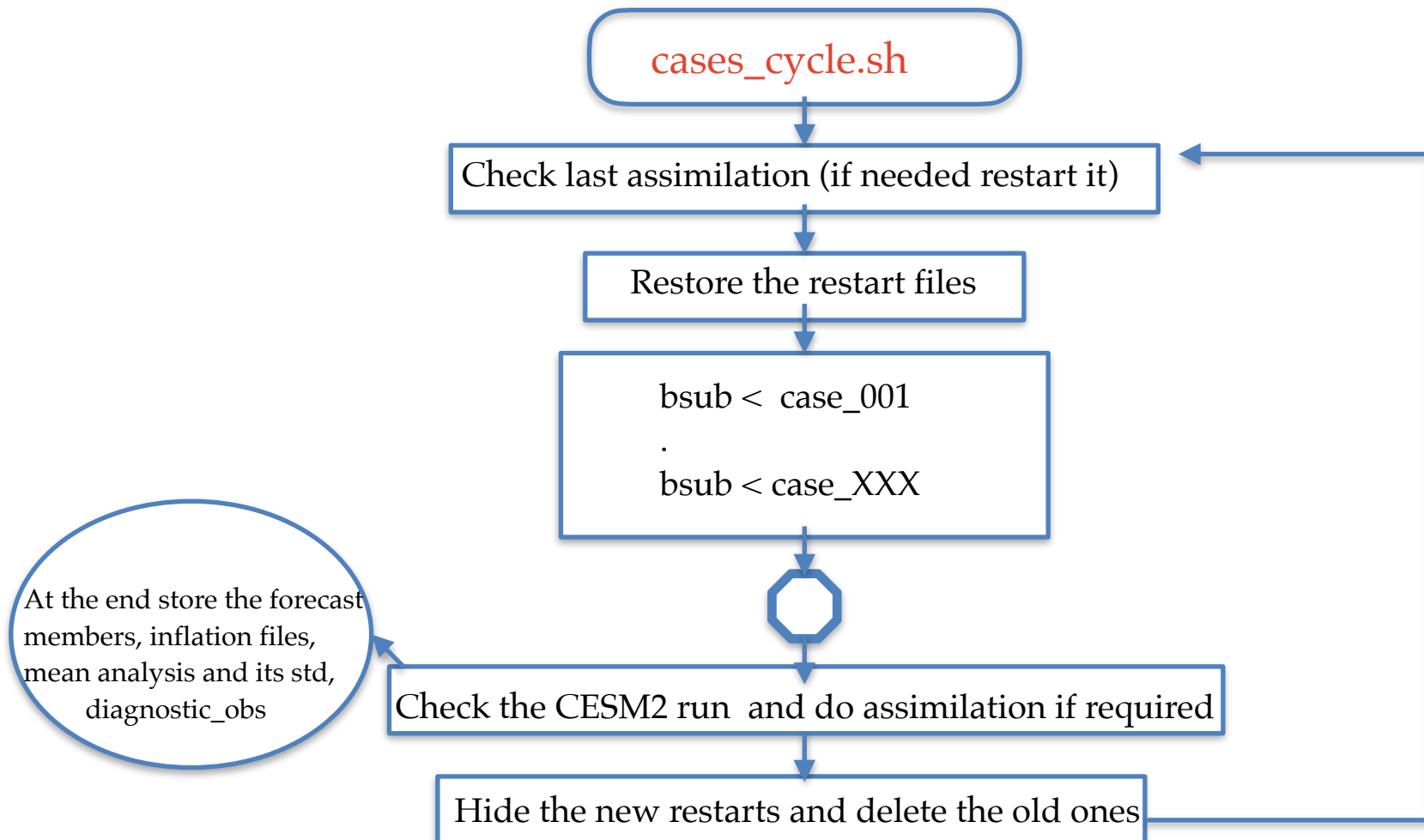
```
cases_cycles.sh  --> cases.assimilate.csh      -
                  |                               | --> cases_check.sh
                  --> cases_no_assimilate.sh    -
                  |
                  --> cases_restart_management.sh
```

cases_check controls if the CESM2 run have been finished correctly.



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Phase 2: RUN THE EXPERIMENT



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Phase 3: DIAGNOSTIC

```
bsub < cases_diag.sh
```

Variable to set:

STARTDATE:	starting date to be examined
ENDDATE:	final date
EXPNAME:	experiment name
TMPROOT:	auxiliary directory where the filter produce output
ARCDIR:	archive dir

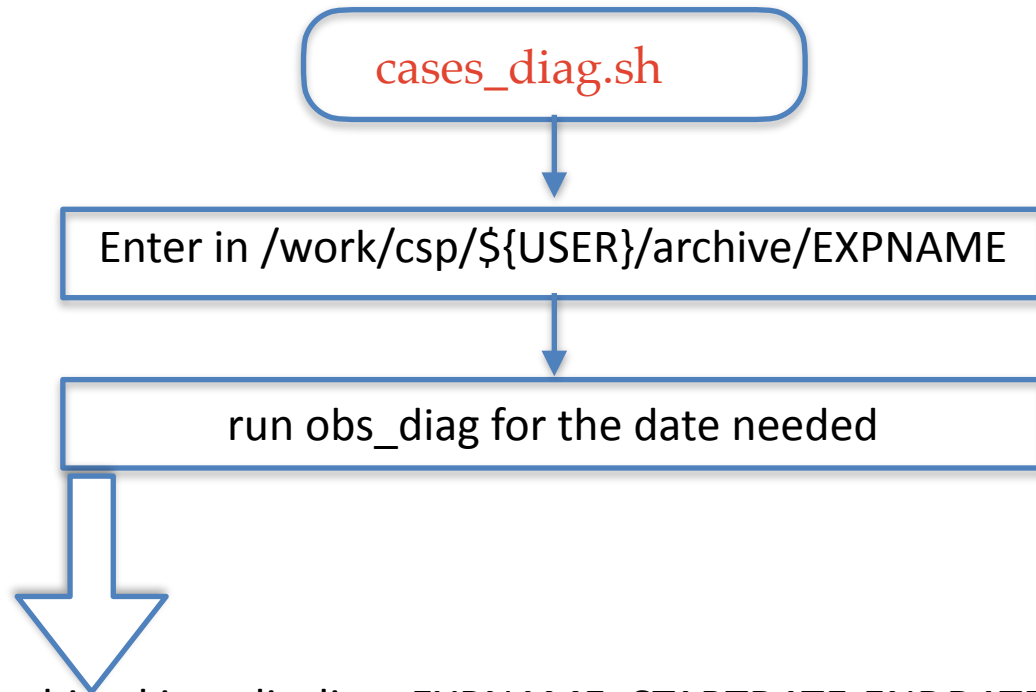
Dependencies: NO

The produced diagnostic file can be then examined with the DART matlab diagnostic tool, or for a quick check with the python tools in the diagnostic/ dir.



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Phase 3: DIAGNOSTIC



The outputs are archived in a dir `diag_EXPNAME_STARTDATE-ENDDATE`



Some results

Just a reminder:

- CAM6 - 0.9x1.25 degree resolution
- 32 levels
- 30 members out of the 80 of NCAR (spin up already performed!)
- CLM5 - Satellite Phenology



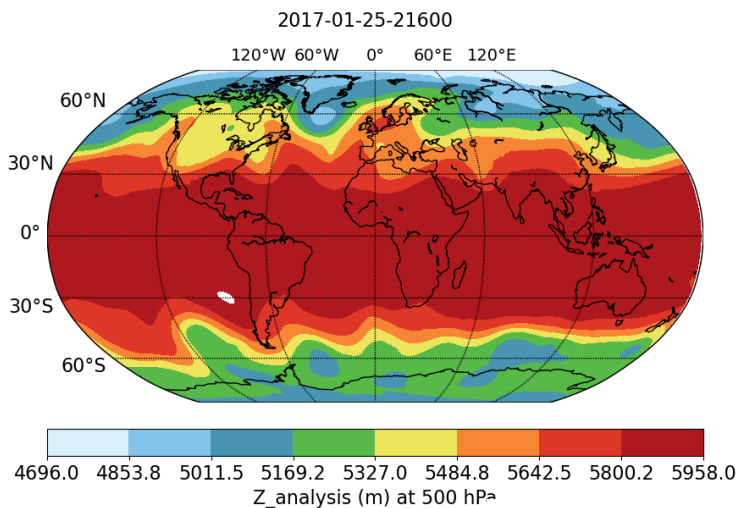
Observations:

1 RADIOSONDE_U_WIND_COMPONENT
2 RADIOSONDE_V_WIND_COMPONENT
5 RADIOSONDE_TEMPERATURE
12 AIRCRAFT_U_WIND_COMPONENT
13 AIRCRAFT_V_WIND_COMPONENT
14 AIRCRAFT_TEMPERATURE
16 ACARS_U_WIND_COMPONENT
17 ACARS_V_WIND_COMPONENT
18 ACARS_TEMPERATURE
30 SAT_U_WIND_COMPONENT
31 SAT_V_WIND_COMPONENT
33 AIRS_TEMPERATURE
143 GPSRO_REFRACTIVITY
6 RADIOSONDE_SPECIFIC_HUMIDITY
34 AIRS_SPECIFIC_HUMIDITY
136 RADIOSONDE_SURFACE_ALTIMETER
138 MARINE_SFC_ALTIMETER
139 LAND_SFC_ALTIMETER

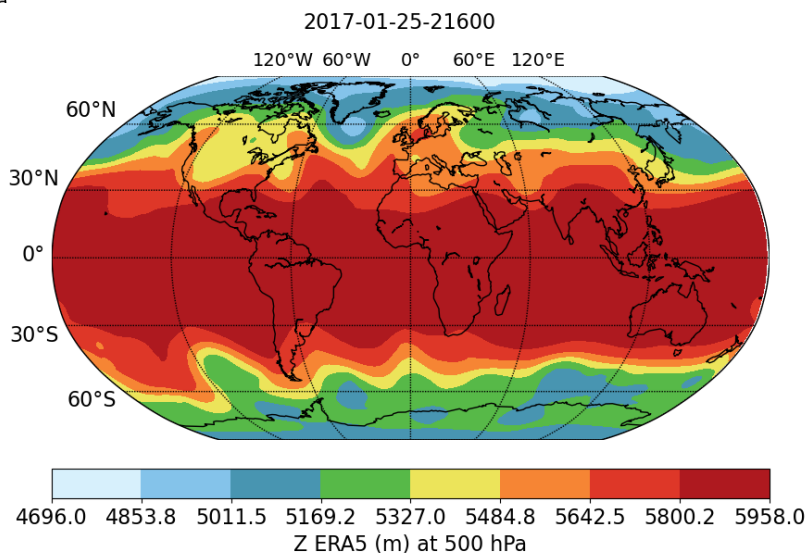
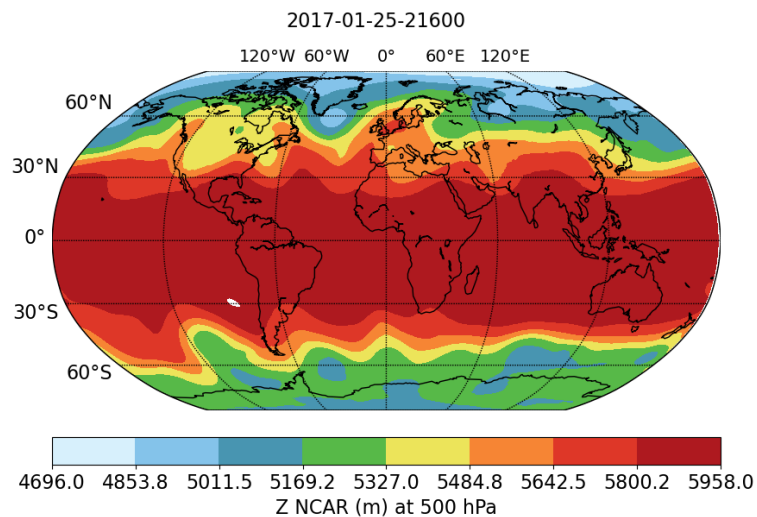


z500:

CMCC Ensemble (30 members)
Kalman Filter



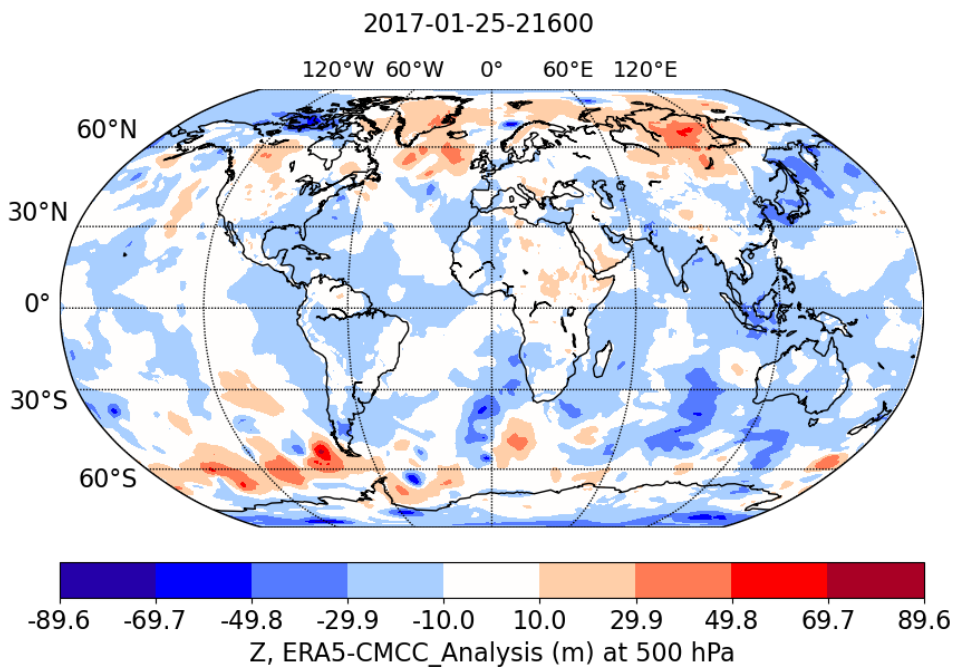
NCAR Ensemble (80 members)
Kalman Filter



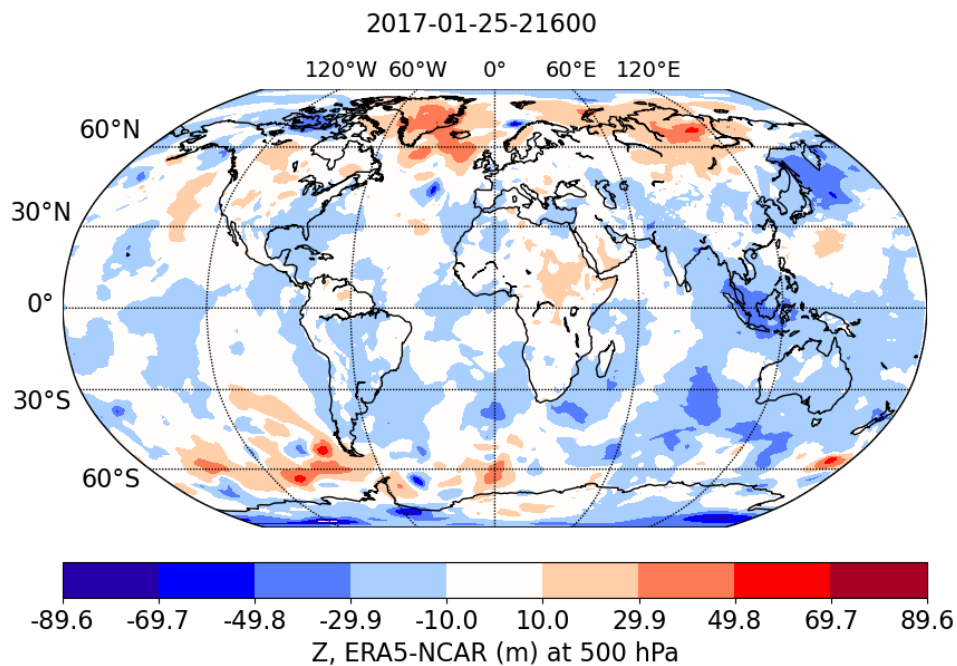
ERA5 (4DVar)



z500 comparison:



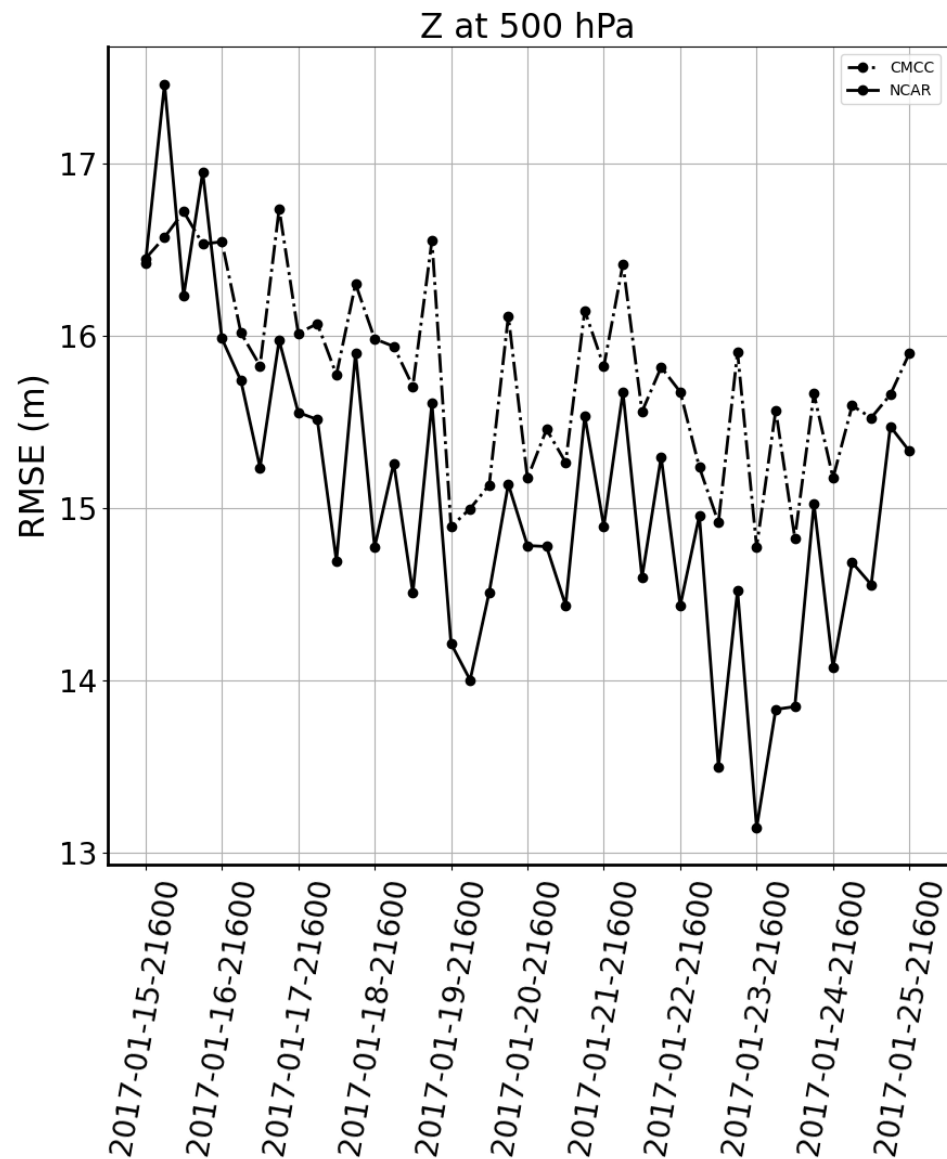
ERA5-CMCC



ERA5-NCAR



z500 comparison: CMCC-NCAR

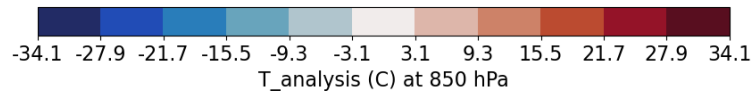
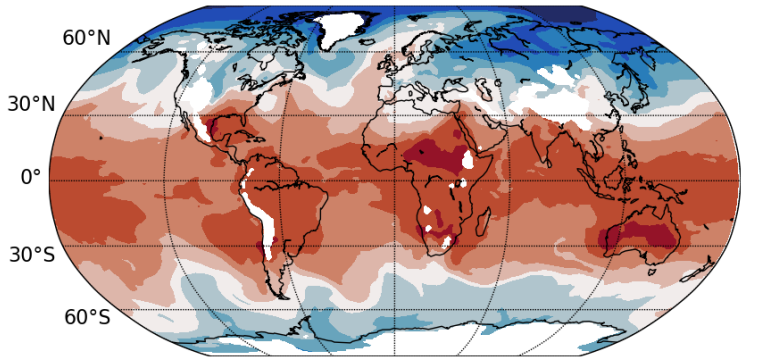


T850:

CMCC Ensemble (30 members) Kalman Filter

2017-01-25-21600

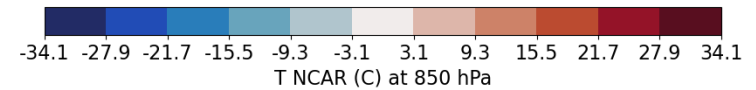
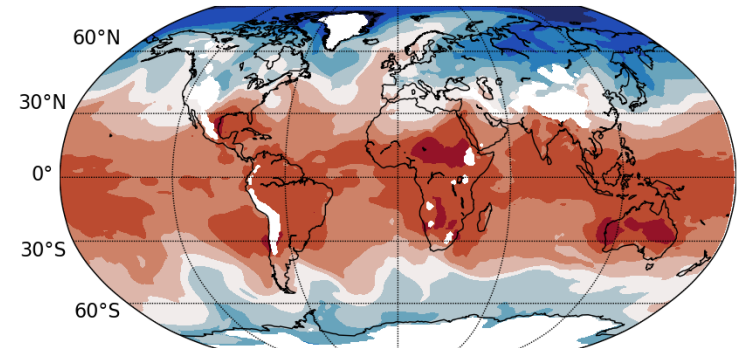
120°W 60°W 0° 60°E 120°E



NCAR Ensemble (80 members) Kalman Filter

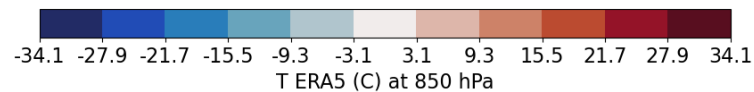
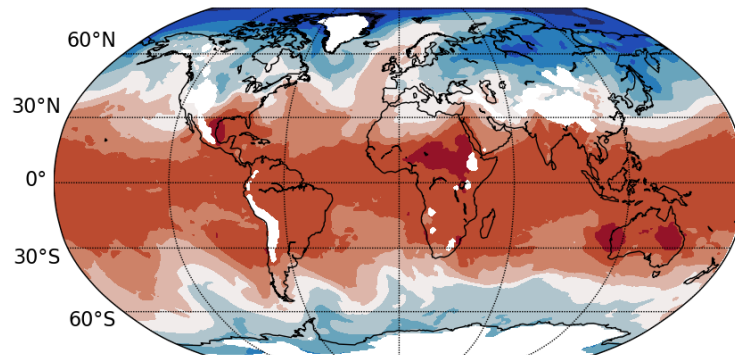
2017-01-25-21600

120°W 60°W 0° 60°E 120°E



2017-01-25-21600

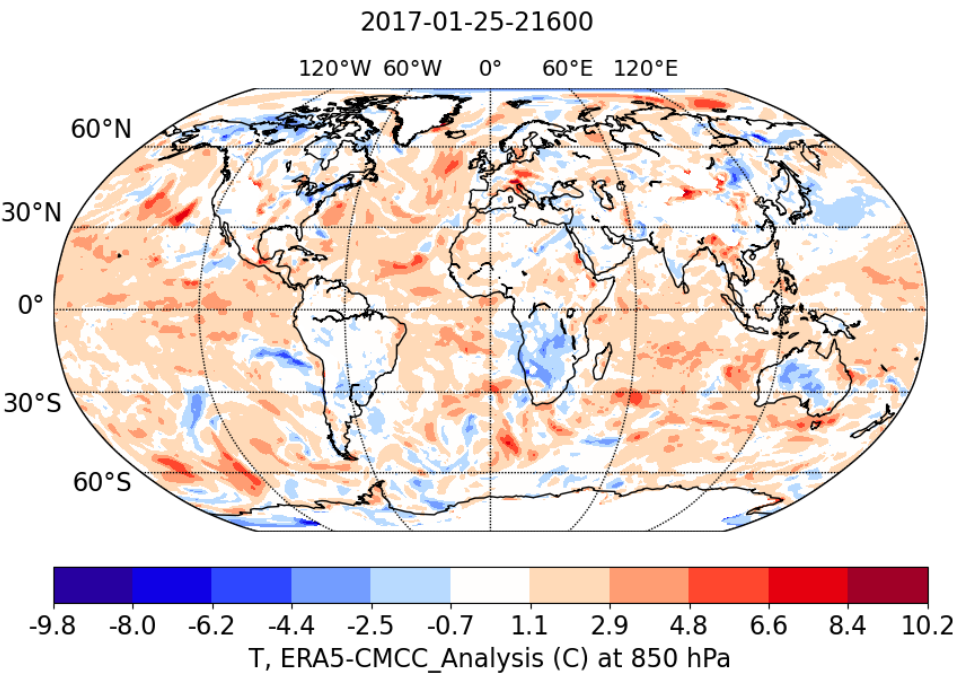
120°W 60°W 0° 60°E 120°E



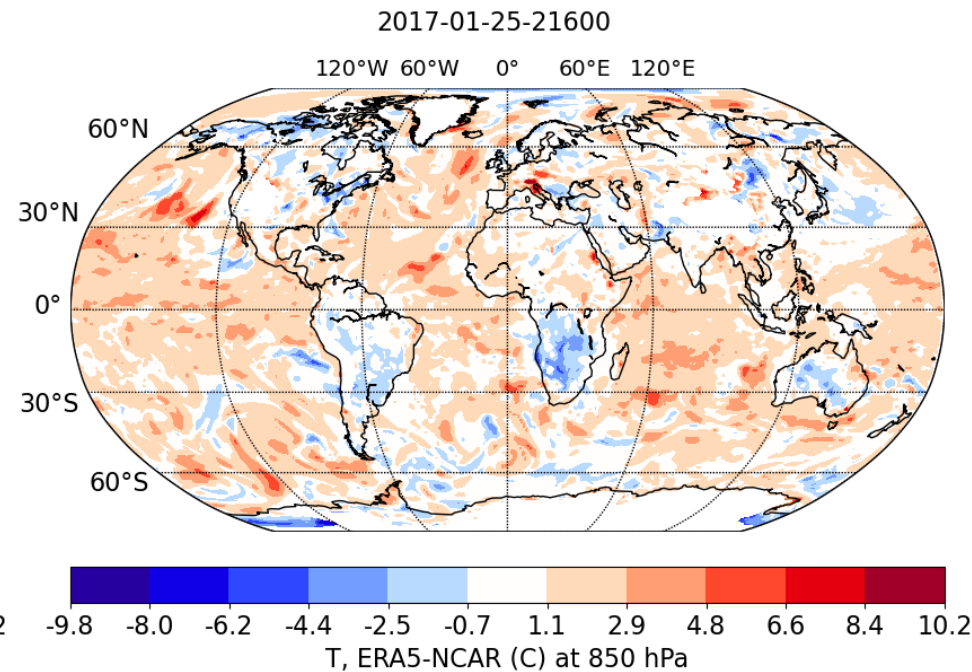
ERA5 (4DVar)



T850 comparison: ERA5-CMCC



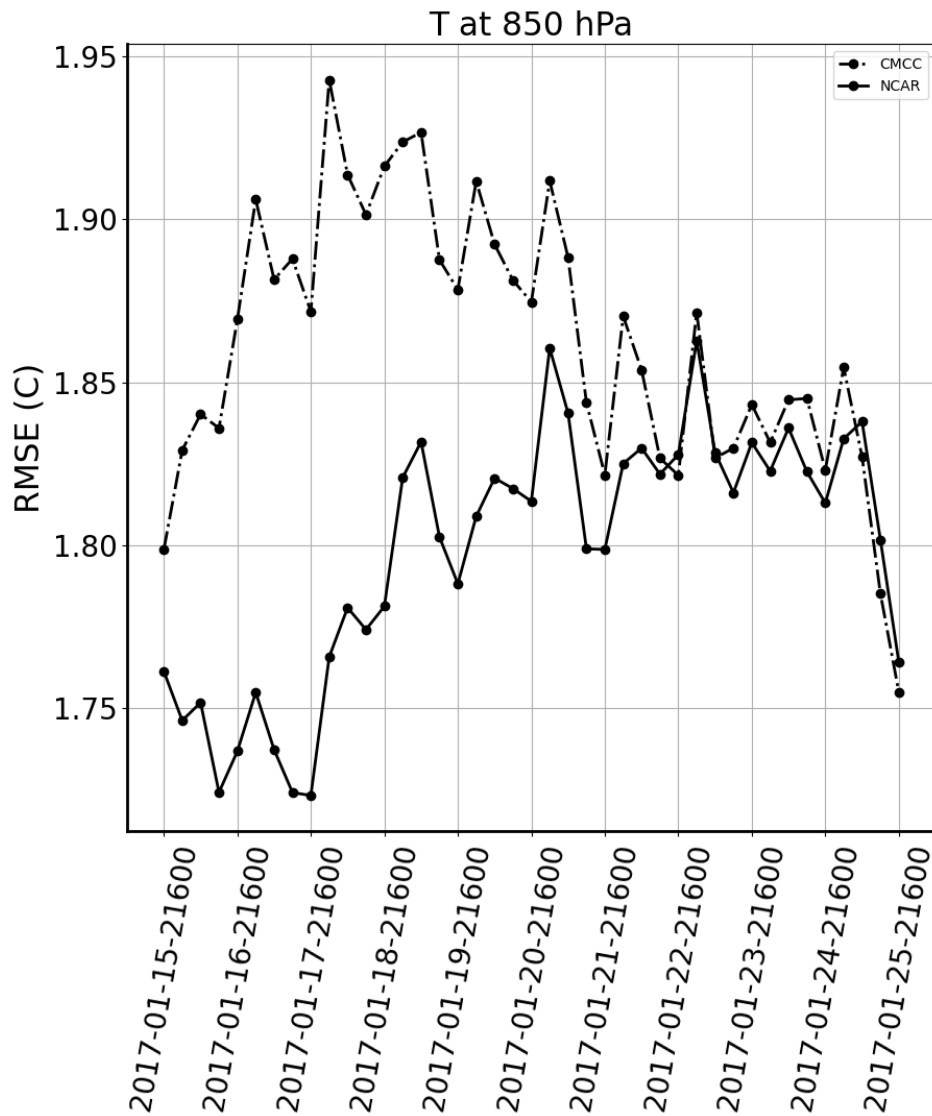
ERA5-CMCC



ERA5-NCAR



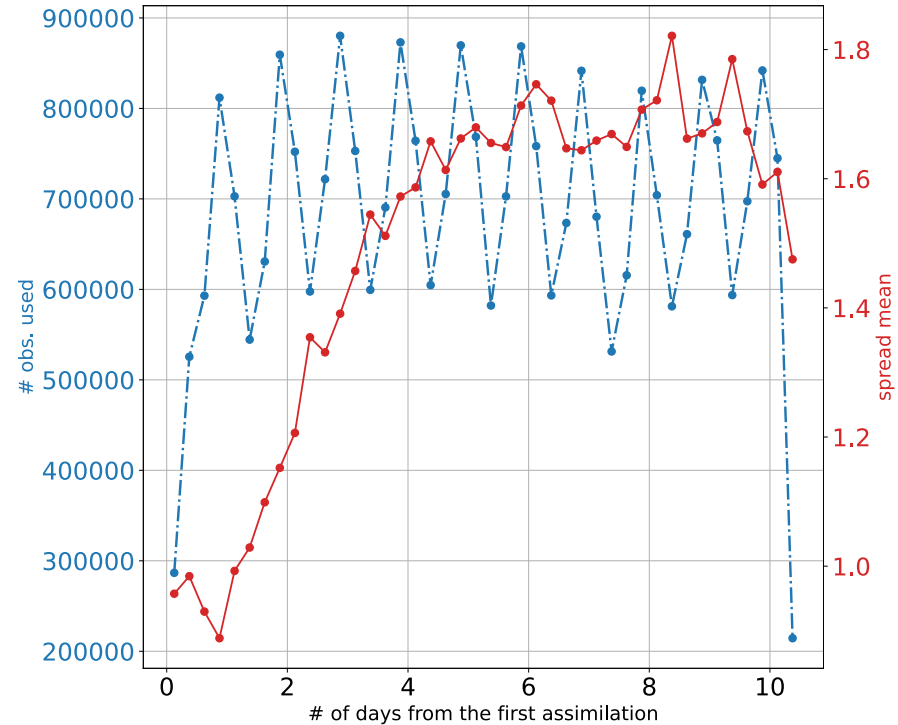
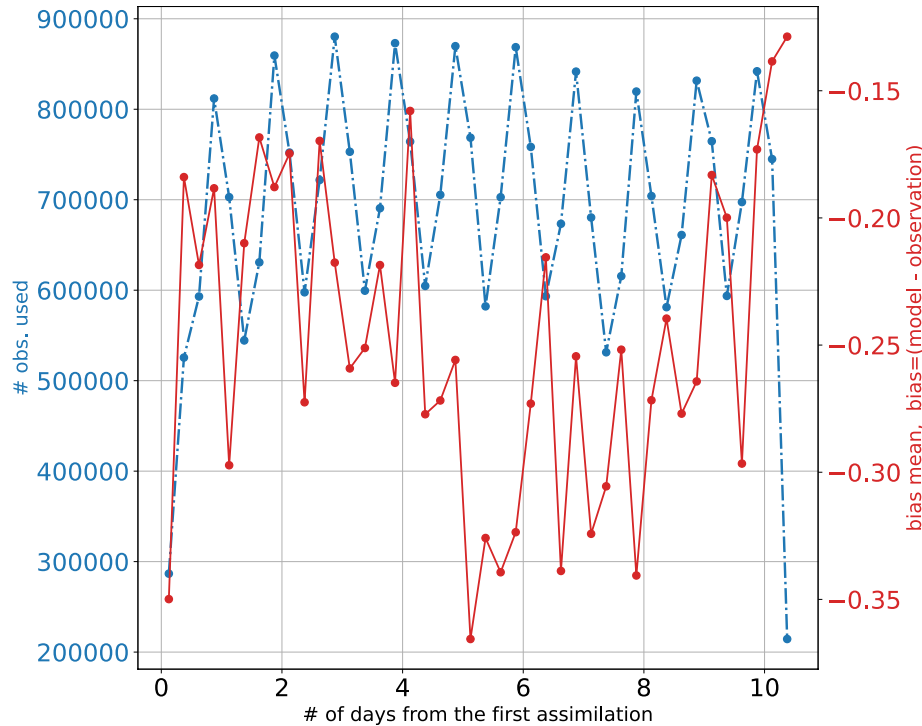
T850 comparison: CMCC-NCAR



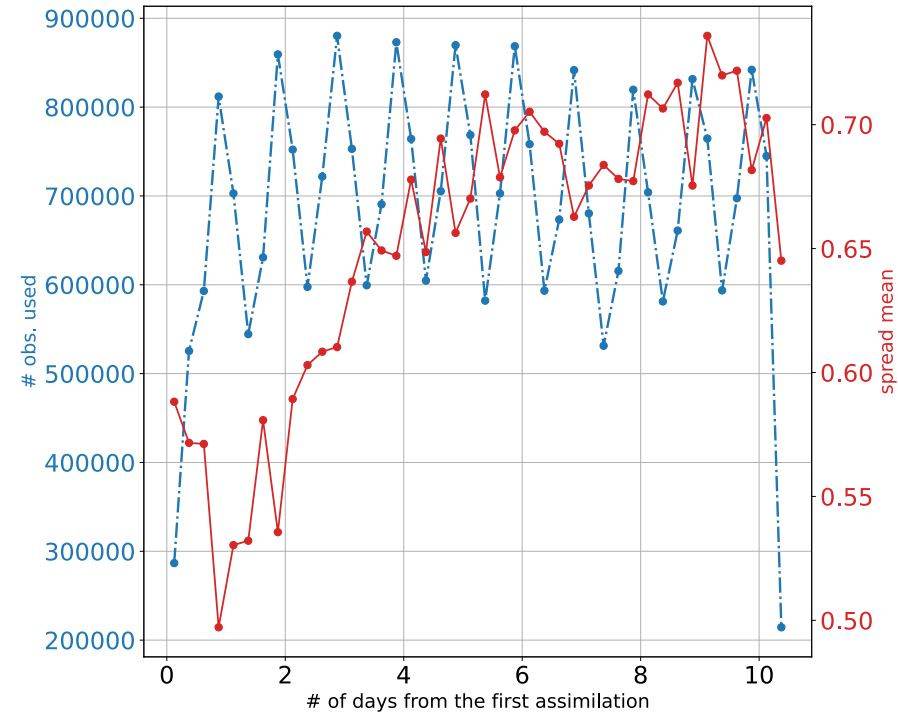
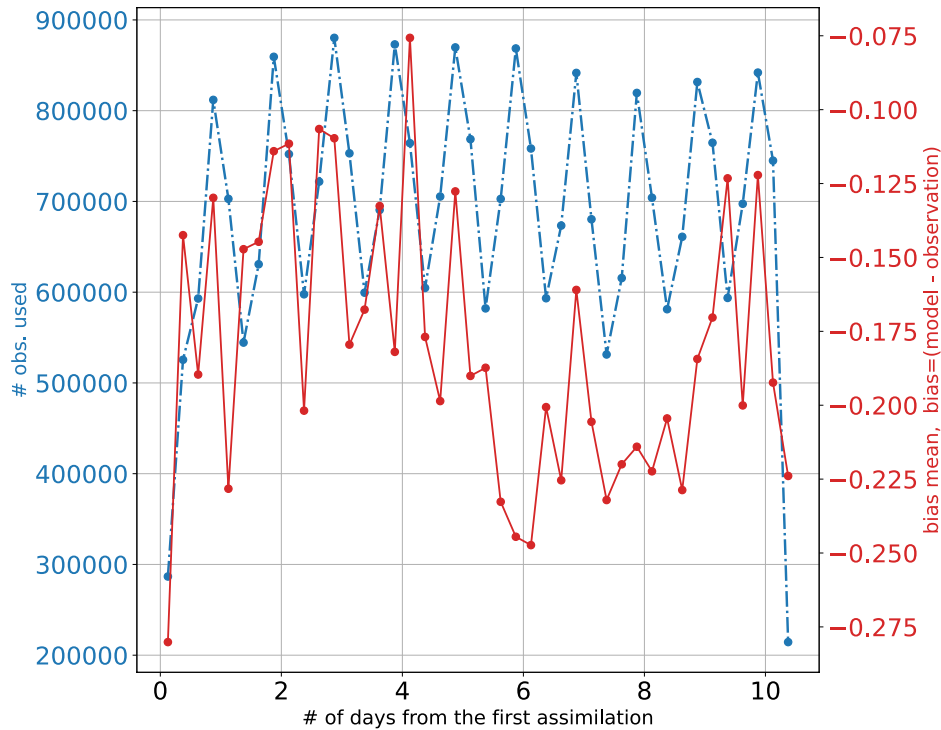
Obs. Space quick diagnostics



Obs. Space quick diagnostics (guess)



Obs. Space quick diagnostics (analysis)



Thanks

